

V	Open Ended Module:		12	
	1	Prepare a report based on Wealth – Income distribution disparities		
	Sections from References:			
Books and References:				
1. Guide to Official Statistics (CSO) 1999				
2. Statistical System in India (CSO) 1995				
3. Principles and Accommodation of National Population Census, UNEDCO.				
4. Monthly Statistics of Foreign Trade in India, DGCIS, Calcutta and other Govt. Publications.				
5. Keyfitz, N (1977): Applied Mathematical Demography- Springer Verlag.				
6. Sen, A(1977): Poverty and Inequality.				
7. Chubey, P.K (1995): Poverty Measurement, New Age International.				

Programme	B. Sc. Statistics				
Course Code	STA5EJ306				
Course Title	Longitudinal Data Analysis				
Type of Course	Major Elective				
Semester	V				
Academic Level	300-399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites					
Course Summary	Learn both how to clean longitudinal data as well as the main statistical models used to analyse it. The course will cover three fundamental frameworks for analysing longitudinal data: multilevel modelling,				

Objective	structural equation modelling and event history analysis.
-----------	---

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Describe the basic concepts of Linear Model in longitudinal data analysis	U	C	Instructor-created exams / Quiz
CO2	Analyze numerical methods to solve the problems in Linear Model	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Explain the basic concepts of Generalized Linear Model	Ap	P	Seminar Presentation / Group Tutorial Work
CO4	Illustrate and study on missing data mechanism in longitudinal data analysis	U	C	Instructor-created exams / Home Assignments
CO5	Analyze longitudinal data using any statistical software	Ap	P	One Minute Reflection Writing assignments
CO6		Ap	P	Viva Voce
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I		Introduction:	12	20
	1	longitudinal studies. Design considerations		
	2	Bias, Efficiency, Sample size calculations.		
	3	Exploring longitudinal data: graphical representation of longitudinal data,.		
	4	fitting smooth curves to longitudinal data,		

	5	Exploring correlation structure.		
	6	General linear models for longitudinal data		
	Sections from References:			
II	Estimation and Analysis		12	15
	7	Weighted least-squares estimation,		
	8	Maximum likelihood estimation. Model-fitting: formulation, estimation, inference.		
	9	Analysis of Variance methods: preliminaries,		
	10	time-by-time ANOVA		
	11	derived variables, repeated measures		
	Sections from References:			
III	Generalized Linear Model		14	20
	12	Generalized Linear Model for Longitudinal Data:		
	13	Marginal models, for binary, ordinal, and count data:		
	14	Random effects models for binary data:		
	15	Random effects models for ordinal data		
	16	Random effects models for count data		
	17	Transition models		
	18	Likelihood-based models for categorical data		
	Sections from References:			
IV			10	15
	19	Dropouts and missing data		
	20	Classification missing data mechanism; Intermittent missing values and dropouts		
	21	Simple solutions and their limitations		
	22	last observation carried forward, complete case analysis		
	Sections from References:			
V	Open Ended Module:		12	